The Avg-Act Swap and Plaintext Overflow Detection in Fully Homomorphic Operations Over Deep Circuits

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Outline

- Introduction
- Definitions
- Challenges and Contributions
- Contribution 1: The Avg-Act Swap
- Contribution 2: Plaintext Overflow Detection
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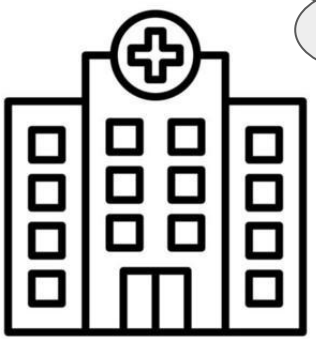
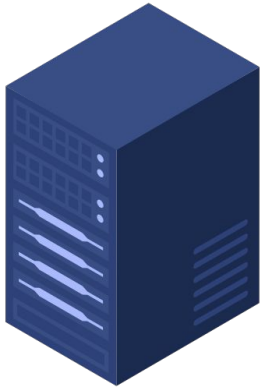
The Scenario in Mind

We want to automatically classify patients' tumor images as benign or malignant.

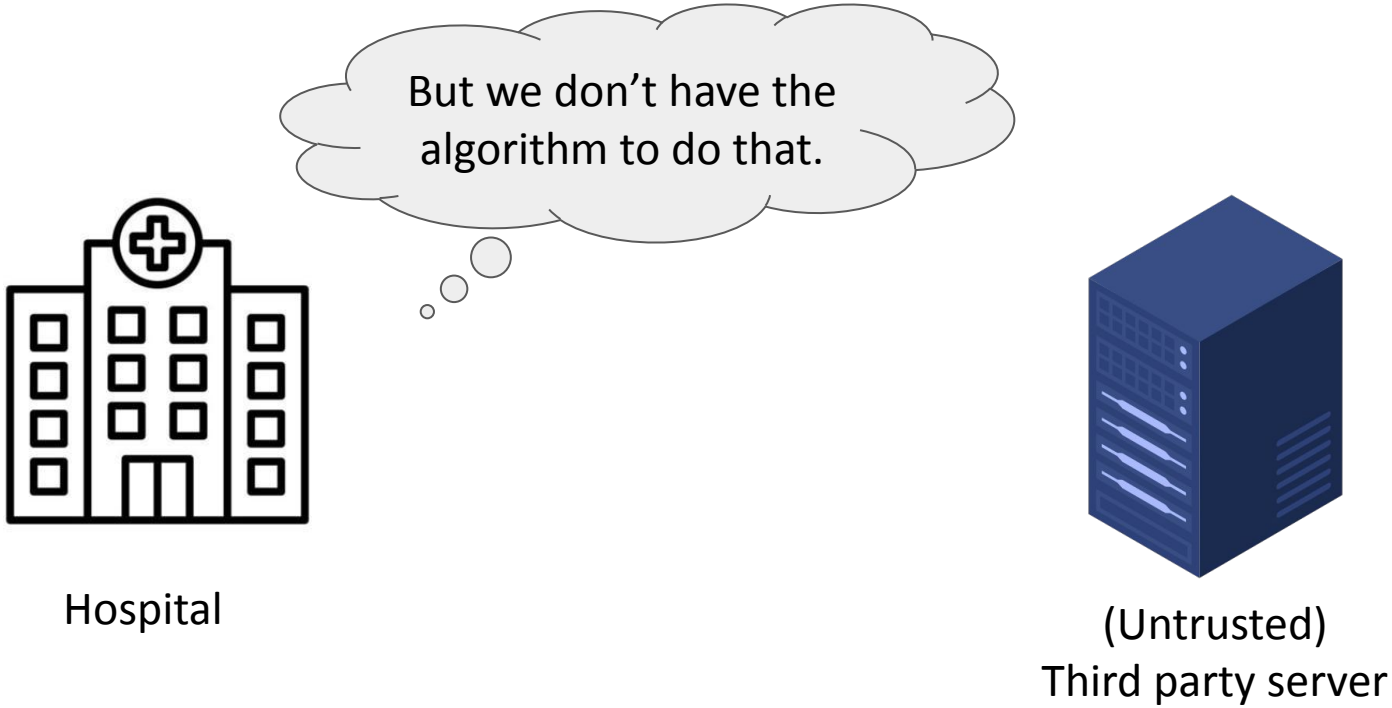


Hospital



(Untrusted)
Third party server

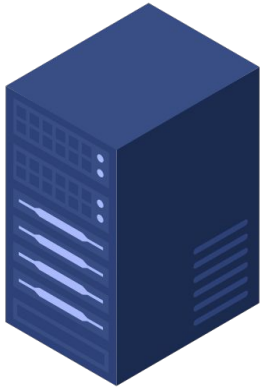
The Scenario in Mind



The Scenario in Mind

Hospital

I have a machine learning algorithm to do that.
Send your images to me.



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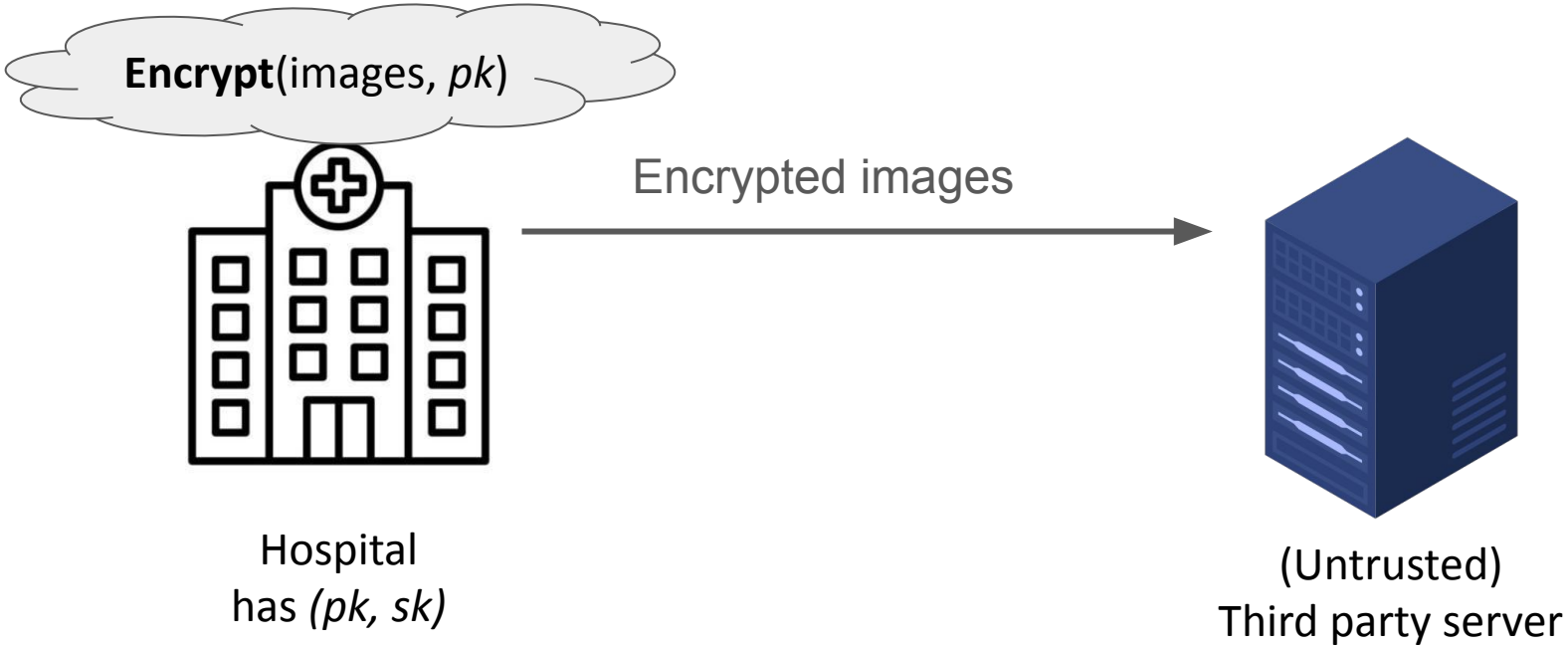
Use Fully Homomorphic Encryption(FHE)!



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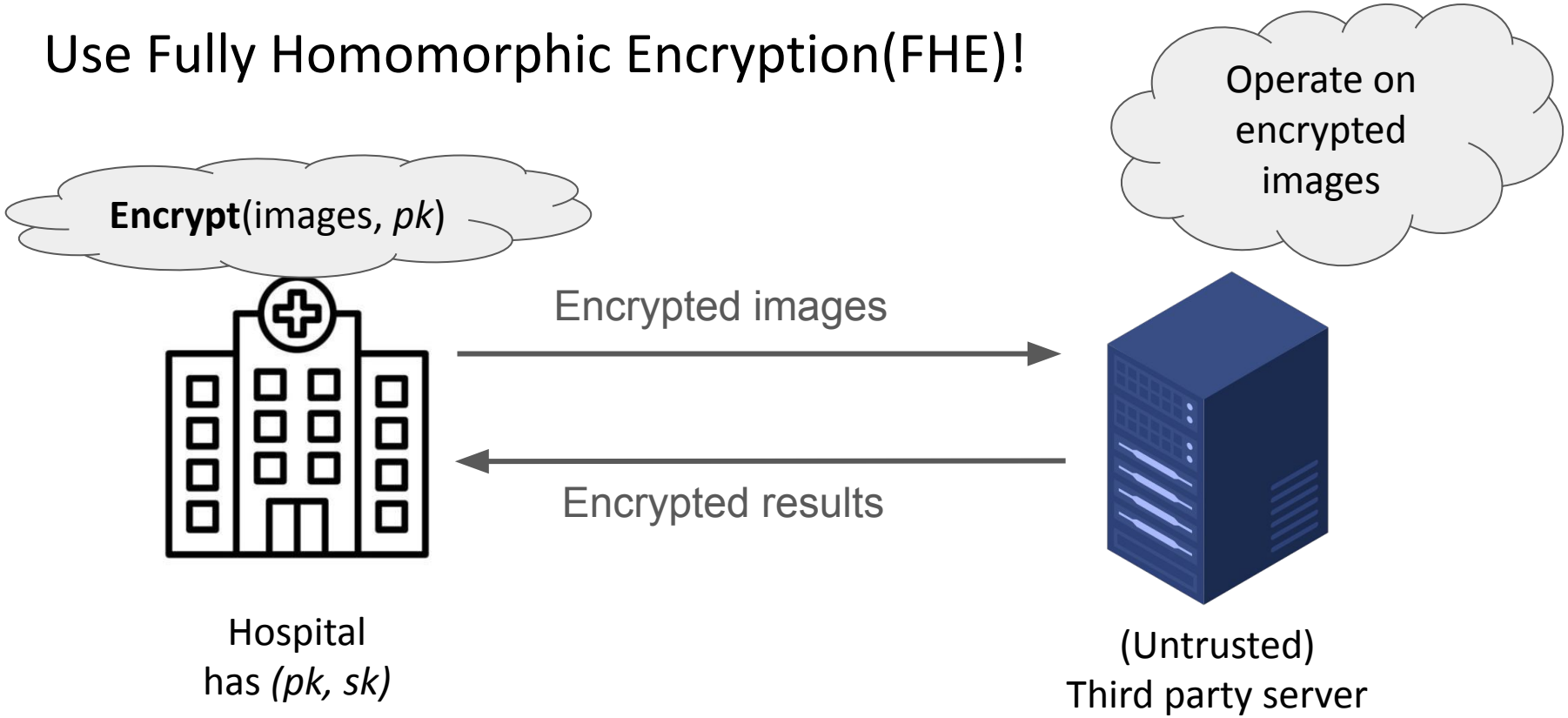

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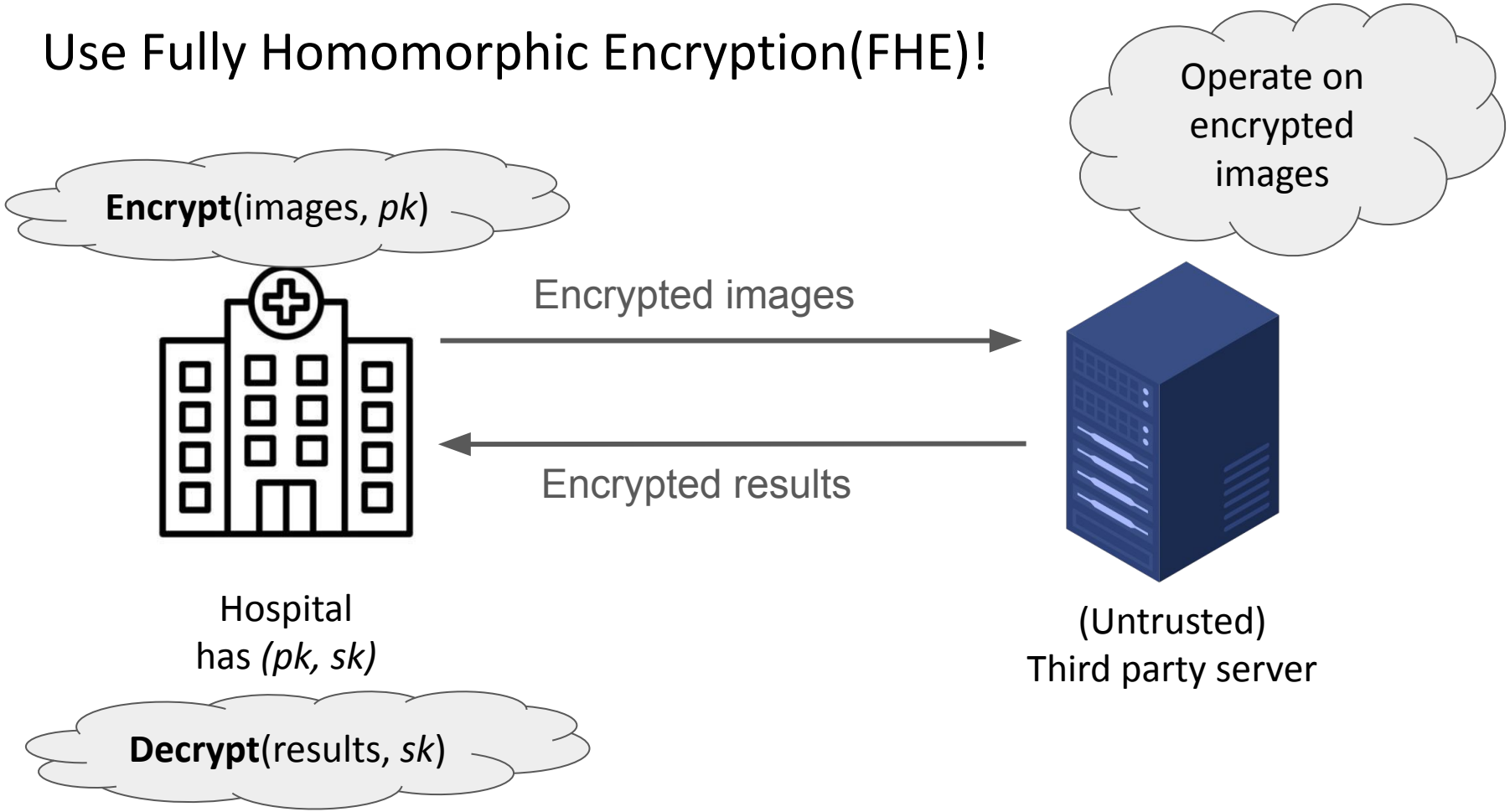

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Fully Homomorphic Encryption (FHE)

- FHE is a public key encryption scheme
- FHE has an efficient algorithm *Evaluate* such that for…
  - valid key pair (pk, sk),
  - ciphertext $ct \leftarrow Encrypt(pk, \pi)$, and
  - any efficient circuit C
- *Evaluate* outputs
  - $\pi' \leftarrow Evaluate(pk, C, ct)$ such that
  - $Decrypt(sk, \pi') = C(\pi)$.

Fully Homomorphic Encryption (FHE)

Consider image classification problems where…

- Circuit *C*: a trained deep neural network
- Ciphertext ct: encryptions of images π
- Evaluation result C(π): the classification of image π

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Challenges and Contributions

- Problem 1: FHE is slow in complex applications like machine learning.
- Contribution 1: **The Avg-Act Swap** reduces encrypted inference time in deep neural networks.

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- Problem 2: Plaintext can overflow after repeated homomorphic operations.
- Contribution 2: **Plaintext overflow detection** in Cheon-Kim-Kim-Song.

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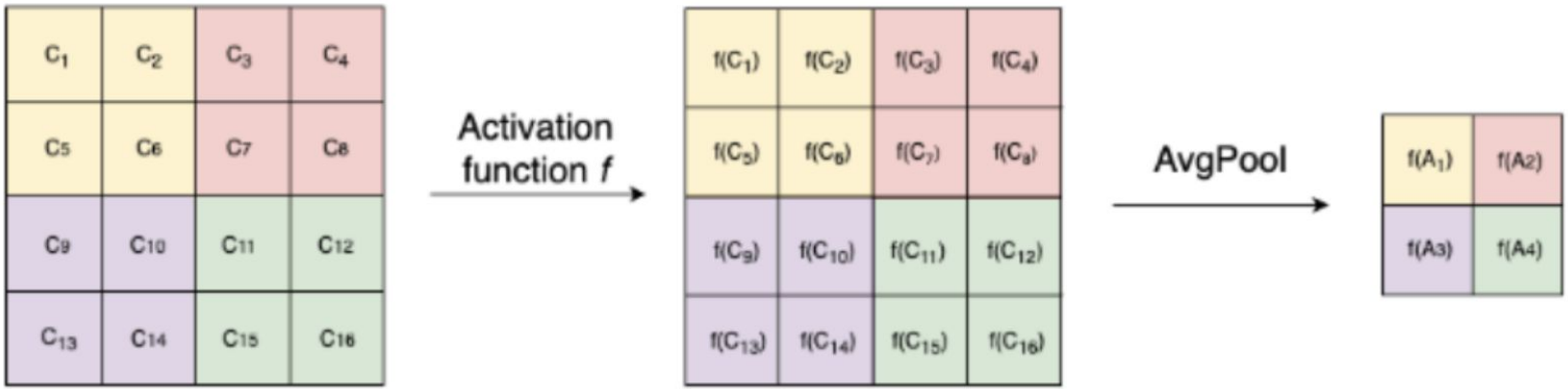
The Avg-Act Swap

In machine learning algorithms for unencrypted data, activation function often comes before AvgPool.



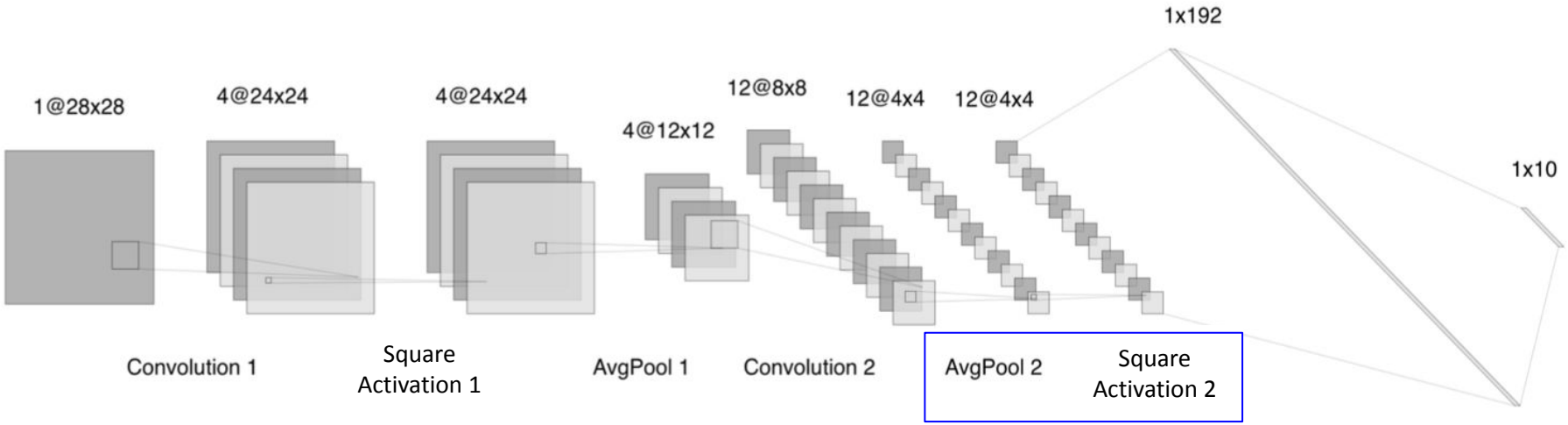
e.g. kernel size = 2

The Avg-Act Swap

However, low speed is the main bottleneck in privacy-preserving machine learning with FHE.

Minimizing Drops in Accuracy

In deeper networks with multiple occurrences of activation-AvgPool pairs, apply the Swap to only the last occurrence.



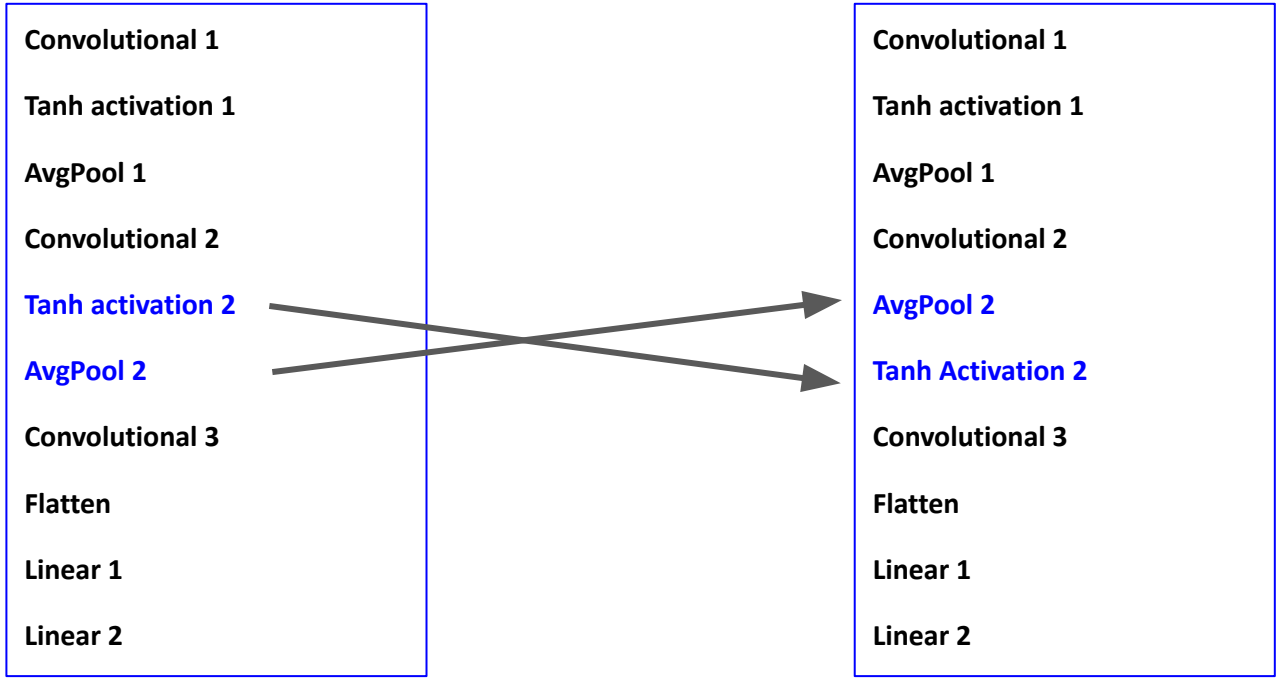
Evaluation

- Trained the network on unencrypted MNIST images
- Tested the model's FHE-friendly equivalent on MNIST test images encrypted with FHE

Kernel size	Total encrypted inference time		
	Trad.	Swap	% decrease
2	945.47	572.60	39.44
3	854.57	572.19	33.04
4	860.62	572.81	33.44
5	848.22	567.75	33.06

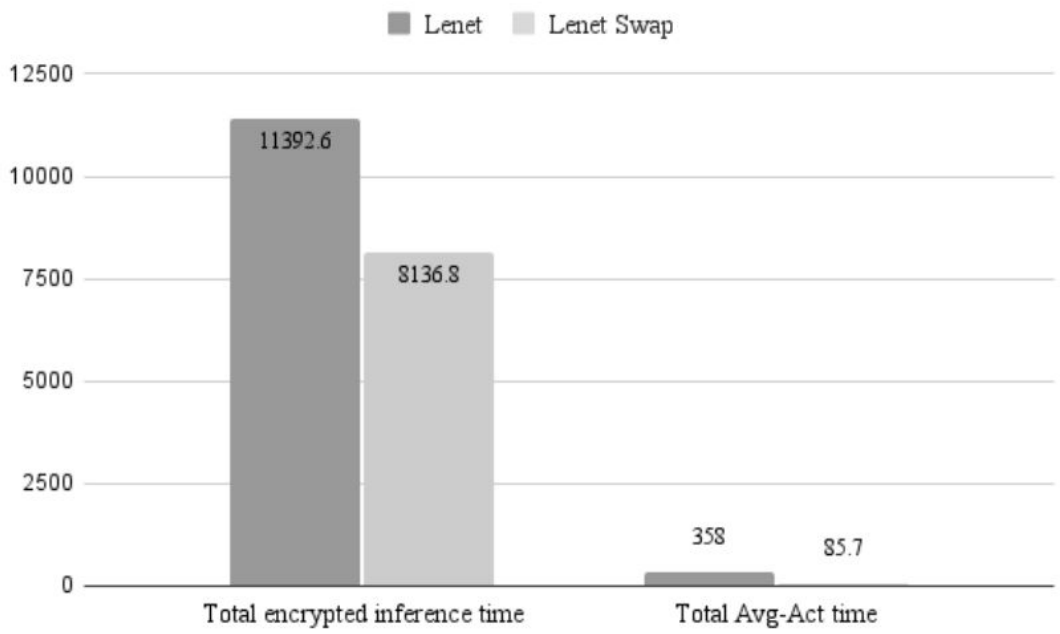
Kernel size	Traditional accuracy			Swap accuracy		
	Plain	FHE	% change	Plain	FHE	% change
2	99	99	0	98.78	98	-1
3	98.02	98	0	97.43	98	0
4	98.50	99	0	97.63	96	-2
5	96.41	97	0	94.63	97	2

Modifying Lenet-5 With the Avg-Act Swap



Swap Tanh activation 2 and AvgPool 2

Evaluation



- 28.6% reduction in encrypted inference time
- 76.1% reduction in Avgpool and Activation time
- 90% accuracy (as compared to 100% without the Avg-Act Swap)

Problem 2: Plaintext Overflow in FHE

- Plaintexts are elements of a ring over a modulus
- Plaintext might grow larger than the modulus
- Decrypted result loses meaning
- How can we detect plaintext overflow *after* all operations are done?

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Plaintext Overflow Detection in Cheon-Kim-Kim-Song (CKKS)

- Every CKKS plaintext is defined by…
  - Noise upper bound: B
  - Plaintext upper bound: v
- Noise bound of a fresh ciphertext: B_{clean}
- Relative error bound β: B/v (proportion of noise contained in plaintext)
- CKKS defines B and β after every operation
- For the actual decrypted result v' after some homomorphic operations, plaintext overflow detected if:

$$\mathbf{B/v' > \beta}$$

Modeled as an Interaction

Client
(owns plaintext)

1. Send encrypted B_{clean} and v

Server
(operates on
encrypted values)

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2. Homomorphically compute

I. The result of operations v'
II. The maximum error bound B_{result}
III. The relative error bound β_{result}

without seeing any underlying plaintexts

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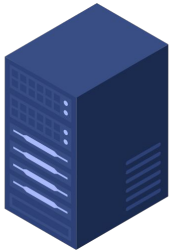
(owns plaintext)

1. Send encrypted B_{clean} and v

3. Send v', B_{result}, β_{result}



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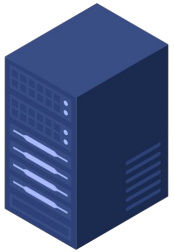

Server
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2. Homomorphically compute
   I. The result of operations v'
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without seeing any underlying plaintexts

4. Decrypt v', B_{result}, β_{result}

If $B_{result}/v' > \beta_{result}$, then plaintext overflowed.

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- **The Avg-Act Swap** increases encrypted inference speed while maintaining accuracy in privacy-preserving machine learning.
- **The plaintext overflow detection scheme** in CKKS helps the client detect whether plaintexts have overflown.

References

- Jung Hee Cheon, Andrey Kim, Miran Kim, and Yongsoo Song. 2017. Homomorphic Encryption for Arithmetic of Approximate Numbers. In Advances in Cryptology – ASIACRYPT 2017, Tsuyoshi Takagi and Thomas Peyrin (Eds.). Springer International Publishing, Cham, 409–437.
- Craig Gentry. 2009. A fully homomorphic encryption scheme. Ph. D. Dissertation. Stanford University.

- Jung Hee Cheon, Andrey Kim, Miran Kim, and Yongsoo Song. 2017. Homomorphic Encryption for Arithmetic of Approximate Numbers. In Advances in Cryptology – ASIACRYPT 2017, Tsuyoshi Takagi and Thomas Peyrin (Eds.). Springer International Publishing, Cham, 409–437.